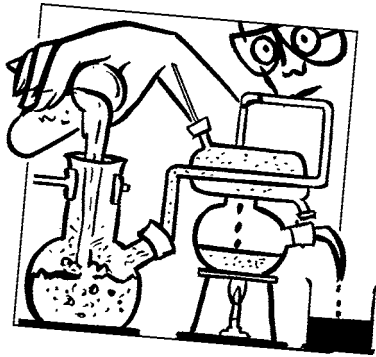


CHAPTER 2

derivatives



The aim of this Chapter...

...is to describe the basic forms of option contracts, make the reader comfortable with the jargon, explain the relevant pages of financial newspapers, give a basic understanding of the purpose of options, and to expand on the 'no free lunch,' or no-arbitrage, idea. By the end of the chapter you will be familiar with the most common forms of derivatives.

In this Chapter...

- the definitions of basic derivative instruments
- option jargon
- no arbitrage and put-call parity
- how to draw payoff diagrams
- simple option strategies

2.1 INTRODUCTION

The previous chapter dealt with some of the basics of financial markets. I didn't go into any detail, just giving the barest outline and setting the scene for this chapter. Here I introduce the theme that is central to the book, the subject of options, a.k.a. derivatives or contingent claims. This chapter is non-technical, being a description of some of the most common option contracts, and explanation of the market-standard jargon. It is in later chapters that I start to get technical.

Options have been around for many years, but it was only on 26th April 1973 that they were first traded on an exchange. It was then that The Chicago Board Options Exchange (CBOE) first created standardized, listed options. Initially there were just calls on 16 stocks. Puts weren't even introduced until 1977. In the US options are traded on CBOE, the American Stock Exchange, the Pacific Stock Exchange and the Philadelphia Stock Exchange. Worldwide, there are over 50 exchanges on which options are traded.

2.2 OPTIONS

If you are reading the book in a linear fashion, from start to finish, then the last topics you read about will have been futures and forwards. The holder of future or forward contracts is *obliged* to trade at the maturity of the contract. Unless the position is closed before maturity the holder must take possession of the commodity, currency or whatever is the subject of the contract, regardless of whether the asset has risen or fallen. Wouldn't it be nice if we only had to take possession of the asset if it had risen?

The simplest **option** gives the holder the *right* to trade in the future at a previously agreed price but takes away the obligation. So if the stock falls, we don't have to buy it after all.

A **call option** is the right to buy a particular asset for an agreed amount at a specified time in the future

As an example, consider the following call option on Microsoft stock. It gives the holder the right to buy one of Microsoft stock for an amount \$25 in one month's time. Today's stock price is \$24.5. The amount '25' which we can pay for the stock is called the **exercise price** or **strike price**. The date on which we must **exercise** our option, if we decide to, is called the **expiry** or **expiration date**. The stock on which the option is based is known as the **underlying asset**.

Let's consider what may happen over the next month, up until expiry. Suppose that nothing happens, that the stock price remains at \$24.5. What do we do at expiry? We could exercise the option, handing over \$25 to receive the stock. Would that be sensible? No, because the stock is only worth \$24.5, either we wouldn't exercise the option or if we really wanted the stock we would buy it in the stock market for the \$24.5. But what if the stock price rises to \$29? Then we'd be laughing, we would exercise the option, paying \$25 for a stock that's worth \$29, a profit of \$4.

We would exercise the option at expiry if the stock is above the strike and not if it is below. If we use S to mean the stock price and E the strike then at expiry the option is worth

$$\max(S - E, 0).$$

This function of the underlying asset is called the **payoff function**. The ‘max’ function represents the optionality.

Why would we buy such an option? Clearly, if you own a call option you want the stock to rise as much as possible. The higher the stock price the greater will be your profit. I will discuss this below, but our decision whether to buy it will depend on how much it costs; the option is valuable, there is no downside to it unlike a future. In our example the option was valued at \$1.875. Where did this number come from? The valuation of options is one of the subjects of this book, and I’ll be showing you how to find this value later on.

What if you believe that the stock is going to fall, is there a contract that you can buy to benefit from the fall in a stock price? Yes, there is.

A **put option** is the right to *sell* a particular asset for an agreed amount at a specified time in the future

The holder of a put option wants the stock price to fall so that he can sell the asset for more than it is worth. The payoff function for a put option is

$$\max(E - S, 0).$$

Now the option is only exercised if the stock falls below the strike price.

Figure 2.1 is an excerpt from *The Wall Street Journal Europe* of 14th April 2005 showing options on various stocks. The table lists closing prices of the underlying stocks and the last traded prices of the options on the stocks. To understand how to read this let us examine the prices of options on Apple. Go to ‘AppleC’ in the list, there are several instances. The closing price on 13th April 2005 was \$41.35 (the LAST column, second from the right). Calls and puts are quoted here with strikes of \$37.50, \$40, . . . , \$47.50, \$50, others may exist but are not included in the newspaper. The expiries mentioned are April, May and July. Part of the information included here is the volume of the transactions in each series; we won’t worry about that but some people use option volume as a trading indicator. From the data, we can see that the April calls with a strike of \$40 were worth \$2.40. The puts with same strike and expiry were worth \$1.20. The April calls with a strike of \$42.50 were worth \$1.20 and the puts with same strike and expiry were worth \$2.45. Note that the higher the strike, the lower the value of the calls but the higher the value of the puts. This makes sense when you remember that the call allows you to buy the underlying for the strike, so that the lower the strike price the more this right is worth to you. The opposite is true for a put since it allows you to sell the underlying for the strike price.

There are more strikes and expiries available for options on indices, so let’s now look at the Index Options section of *The Wall Street Journal Europe* 5th January 2000, this is shown in Figure 2.2.

In Figure 2.3 are the quoted prices of the March and June DJIA calls against the strike price. Also plotted is the payoff function *if the underlying were to finish at its current value at expiry*, the current closing price of the DJIA was 10,997.93.

This plot reinforces the fact that the higher the strike the lower the value of a call option. It also appears that the longer time to maturity the higher the value of the call. Is it obvious that this should be so? As the time to expiry decreases what would we see happen? As

70.72	120,040	June 113.10	113.20	113.06	113.16	0.14	113.20	111.28	1,001,156
7.70	96.21	124,220	Sept 113.07	113.07	113.03	113.11	0.14	113.07	111.83
7.35	96.60	94,895	vol Wed 631,483; open int 1,015,808, +6,768.						
9.95	96.43	37,254							
9.99									
1000; pts 32nds of 100%									
</									

INDEX OPTIONS TRADING									
Tuesday, January 4, 2000									
Volume, close, net change and open interest for all contracts. Volume figures are unofficial. Open interest reflects previous trading day. p-Put. c-Call. The totals for call and put volume and open interest are midday figures.									
Strike	Vol.	3 pm Close	Net. Open Chg.	Int.	Strike	Vol.	3 pm Close	Net. Open Chg.	Int.
CHICAGO									
CB MEXICO INDEX(MEX)									
Jun 90 c	5	199 1/2	- 1 3/8	13	Feb 1500 c	669	12	-12 1/4	5,484
Mar 110 c	10	4 1/4	- 1 1/8	46	Feb 1500 p	55	78 1/2	+ 9	116
Call Vol.	15	Open Int.	116		Mar 1500 c	1,001	32 1/2	- 4 1/2	13,524
Put Vol.	0	Open Int.	338		Mar 1500 p	17	83	+ 8	740
CB TECHNOLOGY(TXX)									
Jan 650 p	10	1/8	- 6 1/8	26	Jan 1525 c	541	3/4	- 3 1/4	5,628
Feb 820 p	10	8 1/8	+ 1 1/8	30	Jan 1525 p	203	97	+ 16	278
Feb 900 p	60	17 1/4	+ 3 1/2	30	Feb 1525 c	181	7 3/4	- 8 1/4	1,861
Call Vol.	0	Open Int.	319		Mar 1525 c	1,606	15	-25 1/2	7,549
Put Vol.	80	Open Int.	381		Mar 1525 p	500	114	+28	218
DJ INDUS AVG(DJX)									
Jan 90 p	540	1/16	...	9,881	Jan 1550 c	789	3/8	- 5/8	13,957
Jan 92 p	180	1/16	...	1,546	Feb 1550 c	221	3 3/8	- 4 7/8	3,059
Feb 92 p	150	3/16	...	150	Feb 1550 p	211	11	+25	77
Mar 92 c	1	21 1/2	- 1	1	Mar 1550 c	1,397	11	- 5 1/2	9,956
Jun 92 p	5	1 1/2	+ 5/16	9,117	Jan 1575 c	10	3/8	+ 1/8	2,102
Jan 96 p	13	1/8	+ 1/16	23,136	Feb 1575 c	6	2	- 3 3/8	1,779
Feb 96 p	23	3/16	+ 1/4	1,340	Mar 1575 c	209	6 3/4	- 6	619
Mar 96 p	50	1	+ 3/16	10,585	Mar 1575 p	10	152 1/2	+38 1/2	11
Jun 96 p	795	1 7/8	+ 1/4	6,974	Feb 1600 c	155	1	- 1 1/8	1,925
Jan 100 p	35	5/16	+ 3/16	9,841	Mar 1600 c	502	3 3/4	- 3 1/4	9,199
Feb 100 p	112	1 1/16	+ 1/4	2,983	Mar 1650 c	55	1 1/4	- 1 3/4	2,052
Mar 100 c	82	13 1/4	- 3 3/4	258	Jan 1700 p	1	267	+ 15	373
Jun 100 p	5	2 3/4	+ 3/4	5,023	Mar 1700 c	100	1	- 1 1/8	1,416
Jan 102 p	1,088	5/16	+ 1/8	4,358	Mar 1700 p	3	264	+22	32
Feb 102 p	365	1 1/16	+ 1/4	1,750	Call Vol.	36,803	Open Int.	820,987	
Mar 102 p	30	1 1/16	+ 3/8	3,147	Put Vol.	45,134	Open Int.	975,013	
Jan 104 c	111	9 1/8	- 7/8	2,654	AMERICAN				
Jan 104 p	449	1/2	+ 1/4	3,671	COMP TECH(XCI)				
Feb 104 p	4	1 1/16	...	227	Jan 860 p	3	16 1/4	...	...
Mar 104 c	2	10 1/4	- 1 1/8	1,051	Jan 1120 c	22	259 3/4	+160 1/8	22
Mar 104 p	8	1 1/8	+ 3/8	2,367	Jan 1170 p	3	7	+ 2 7/8	3
Jun 104 p	5	3 1/2	+ 3/4	1,632	Call Vol.	22	Open Int.	71	
Jan 105 p	1,281	3/4	+ 3/8	959	Put Vol.	6	Open Int.	60	
Jan 106 c	10	7 1/8	- 1	377	JAPAN INDEX(JPN)				
Jan 106 p	28	3/4	+ 3/8	1,999	Mar 170 c	2	25 1/4	+ 3 3/4	651
Mar 106 c	8	9 3/8	- 2 1/4	783	Mar 175 c	9	21	- 1	39
Mar 106 p	1	2 5/16	+ 5/16	757	Jan 180 p	30	1 1/16	- 1/8	125
Jun 106 c	2	12 1/2	+ 1/8	3	Mar 180 c	8	17 1/2	+ 3 1/2	3,390
Jan 108 c	23	4 1/2	- 3 3/8	618	Jan 185 c	10	9 5/8	+ 1/8	10
Jan 108 p	159	1 1/8	+ 1/16	2,157	Jan 185 p	10	1 1/16	+ 1/4	91
Feb 108 c	25	2 1/2	+ 5/8	272	Feb 185 p	10	3	- 3/4	10
Mar 108 c	8	7 3/8	- 3	3,594	Mar 185 c	1	14 3/8	- 1 1/4	99
Mar 108 p	8	3 1/8	+ 1/16	4,769	Jan 190 p	35	2 1/8	+ 1/16	145
Jun 108 c	12	10 1/4	- 2 3/4	311	Mar 190 c	156	10 3/4	- 1 1/2	252
Jan 109 c	6	3 3/8	- 3 3/8	30	Jan 195 c	1	3 3/8	- 7/8	42
Jan 109 p	43	1 1/2	+ 1 1/16	689	Feb 195 p	35	5 3/4	- 4 1/8	39
Jan 110 c	143	2 3/4	- 3 1/8	1,519	Mar 195 c	4	8 1/4	- 1/8	101
Jan 110 p	213	1 1/16	+ 1 1/16	5,035	Call Vol.	556	Open Int.	23,806	
Feb 110 c	13	4 5/8	- 2	63	Put Vol.	120	Open Int.	7,561	
Feb 110 p	58	3	+ 1 1/16	514	MS CYCLICAL(CYC)				
Mar 110 c	34	5 1/8	- 2 5/8	7,029	Jan 520 p	125	3 1/8	...	125
					Feb 520 p	2,000	7 1/2	- 1 1/2	5,000
					Feb 560 c	2,000	22	- 4	5,000
					Jan 590 c	20	4 1/2	- 7 1/2	400
					Call Vol.	2,020	Open Int.	12,676	
					Put Vol.	2,125	Open Int.	11,224	
					MS HITECH 35(MSH)				
					Jan 1560 c	50	310 1/4	+10 1/4	39
					Jan 1570 c	850	303 3/4	-36 1/4	865
					Jan 1580 c	50	291 1/4	+23 1/2	14
					Feb 1600 p	275	45	+10	260
					Jan 1610 p	60	14 3/4	- 8 1/8	21
					Jan 1630 c	4	218 3/4	-11 1/8	3

Figure 2.2 The Wall Street Journal Europe of 5th January 2000, Index Options. Reproduced by permission of Dow Jones & Company, Inc.

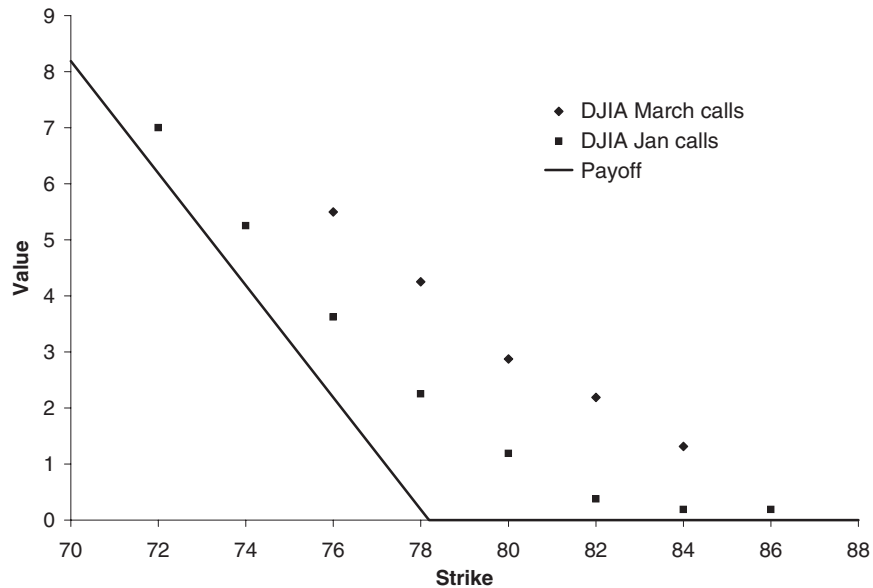


Figure 2.3 Option prices versus strike, March and June series of DJIA.

there is less and less time for the underlying to move, so the option value must converge to the payoff function.



Time Out...

Plotting

When plotting using Excel you'll find it best to use the 'XY Scatter' option. This allows you to get the correct scale on the horizontal axis without any hassle. Also, don't use the smoothing option as it can give spurious wiggles in the plots.

One of the most interesting features of calls and puts is that they have a non-linear dependence on the underlying asset. This contrasts with futures which have a linear dependence on the underlying. This non-linearity is very important in the pricing of options, as the randomness in the underlying asset and the curvature of the option value with respect to the asset are intimately related.

Calls and puts are the two simplest forms of option. For this reason they are often referred to as **vanilla** because of the ubiquity of that flavor. There are many, many more kinds of options, some of which will be described and examined later on. Other terms

GLXO LN GBp ↑ 1688 -13 L 5s L 1686/1689 L Trd Equity OCM At 12:50 Vol 854,194 Op 1694 L Hi 1703 L Lo 1686 L Prev 1701 OPTION MONITOR 3 COMP Center: 1687 1 <GO> to Edit Spreadsheet													
	BID	ASK	LAST	mICHG	mIVBD	IVAS	BEST	DEBS	GABS	VEBS	THEO	7DEC	
GLXO LN	Bid	Ask	Last	1 Day	Imp	Imp	Delta	Gamma	Vega				
CALLS	Price	Price	Trade	Change	Bid	Ask	Best	Best	Best	Best	Theo.	7 Day	
	Price	Price					Price	Price	Price	Price	Value	Decay	
GLXOCT99	1686.0	1689.0	1688.0	-13.0			1687						
1)	1200	489.50	504.50	509.50	unch	N.A.	69.97	504.50	.942	.0003	.674	494.094	.6870
2)	1250	440.00	455.00	460.00	unch	N.A.	63.58	455.00	.936	.0003	.689	444.924	.6853
3)	1300	390.50	405.50	410.50	unch	N.A.	57.36	405.50	.928	.0004	.837	396.334	.6828
4)	1350	342.00	357.00	362.00	unch	N.A.	52.29	357.00	.915	.0005	.853	348.724	.8888
5)	1400	294.50	309.50	314.50	unch	N.A.	48.07	309.50	.895	.0007	1.018	302.625	.2385
6)	1450	249.00	264.00	268.50	unch	29.45	45.11	264.00	.864	.0008	1.194	258.665	.8316
7)	1500	203.00	218.00	224.00	unch	30.67	42.27	220.00	.823	.0011	1.538	217.536	.3538
8)	1600	125.00	137.50	136.00	-6.00	29.86	37.59	136.00	.706	.0017	2.013	146.027	.0423
9)	1700	69.00	76.00	80.00	unch	30.95	34.02	76.00	.516	.0020	2.280	90.956	.4785
10)	1800	32.00	38.00	40.00	unch	30.62	33.12	37.00	.319	.0019	2.005	52.713	.2390
11)	1900	16.00	20.00	21.50	unch	32.84	35.47	20.00	.190	.0013	1.552	28.386	.4811
12)	2000	6.00	9.00	9.00	unch	32.53	35.83	9.00	.099	.0008	1.041	14.273	.2966
13)	2100	2.00	4.00	3.50	unch	32.32	36.52	3.50	.044	.0005	.581	6.728	.1456
14)	2200		2.00	1.00	unch	N.A.	38.08	1.00	.015	.0002	.232	2.968	.5272
15)	2300		1.50	.50	unch	N.A.	41.58	.50	.008	.0001	.132	1.262	.2929
16)	2400		1.00	.50	unch	N.A.	43.98	.50	.007	.0001	.126	.502	.2977
17)	2500		1.00	.50	unch	N.A.	48.40	.50	.007	.0001	.101	.195	.3010

Copyright 1999 BLOOMBERG L.P. Frankfurt: 69-920410 Hong Kong: 2-977-6000 London: 171-330-7500 New York: 212-318-2000
 Princeton: 609-279-3000 Singapore: 226-3000 Sydney: 2-9777-8686 Tokyo: 3-3201-8900 Sao Paulo: 11-3048-4500
 1574-414-0 08-Sep-99 11:50:14

Bloomberg
 PROFESSIONAL

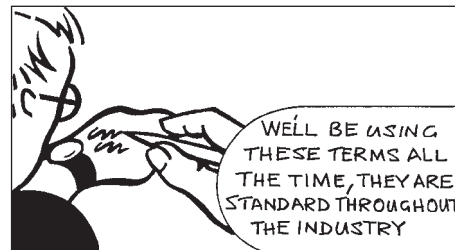
Figure 2.4 Prices for Glaxo–Wellcome calls expiring in October. Source: Bloomberg L.P.

used to describe contracts with some dependence on a more fundamental asset are **derivatives** or **contingent claims**.

Figure 2.4 shows the prices of call options on Glaxo–Wellcome for a variety of strikes. All these options are expiring in October. The table shows many other quantities that we will be seeing later on.

2.3 DEFINITION OF COMMON TERMS

The subjects of mathematical finance and derivatives theory are filled with jargon. The jargon comes from both the mathematical world and the financial world. Generally speaking the jargon from finance is aimed at simplifying communication, and to put everyone on the same footing.¹ Here are a few loose definitions to be going on with, some you have already seen and there will be many more throughout the book.



- **Premium:** The amount paid for the contract initially. How to find this value is the subject of much of this book.
- **Underlying (asset):** The financial instrument on which the option value depends. Stocks, commodities, currencies and indices are going to be denoted by S . The option payoff is defined as some function of the underlying asset at expiry.

¹ I have serious doubts about the purpose of most of the math jargon.

- **Strike (price) or exercise price:** The amount for which the underlying can be bought (call) or sold (put). This will be denoted by E . This definition only really applies to the simple calls and puts. We will see more complicated contracts in later chapters and the definition of strike or exercise price will be extended.
- **Expiration (date) or expiry (date):** Date on which the option can be exercised or date on which the option ceases to exist or give the holder any rights. This will be denoted by T .
- **Intrinsic value:** The payoff that would be received if the underlying is at its current level when the option expires.
- **Time value:** Any value that the option has above its intrinsic value. The uncertainty surrounding the future value of the underlying asset means that the option value is generally different from the intrinsic value.
- **In the money:** An option with positive intrinsic value. A call option when the asset price is above the strike, a put option when the asset price is below the strike.
- **Out of the money:** An option with no intrinsic value, only time value. A call option when the asset price is below the strike, a put option when the asset price is above the strike.
- **At the money:** A call or put with a strike that is close to the current asset level.
- **Long position:** A positive amount of a quantity, or a positive exposure to a quantity.
- **Short position:** A negative amount of a quantity, or a negative exposure to a quantity. Many assets can be sold short, with some constraints on the length of time before they must be bought back.

2.4 PAYOFF DIAGRAMS

The understanding of options is helped by the visual interpretation of an option's value at expiry. We can plot the value of an option at expiry as a function of the underlying in what is known as a **payoff diagram**. At expiry the option is worth a known amount. In the case of a call option the contract is worth $\max(S - E, 0)$. This function is the bold line in Figure 2.5.

Figure 2.6 shows Bloomberg's standard option valuation screen and Figure 2.7 shows the value against the underlying and the payoff.

The payoff for a put option is $\max(E - S, 0)$; this is the bold line plotted in Figure 2.8.

Figure 2.9 shows Bloomberg's option valuation screen and Figure 2.10 shows the value against the underlying and the payoff.

These payoff diagrams are useful since they simplify the analysis of complex strategies involving more than one option.

Make a mental note of the thin lines in all of these figures. The meaning of these will be explained very shortly.

2.4.1 Other representations of value

The payoff diagrams shown above only tell you about what happens at expiry, how much money your option contract is worth at that time. It makes no allowance for how much

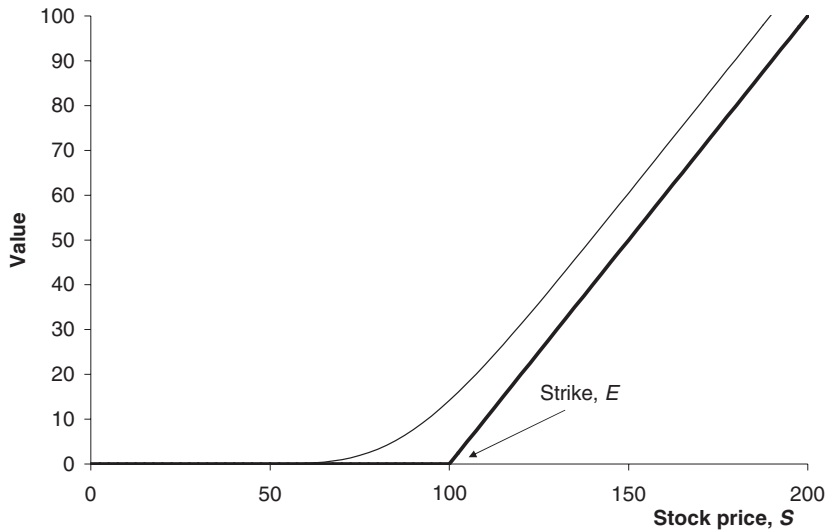


Figure 2.5 Payoff diagram for a call option.

<HELP> for explanation, <MENU> for similar functions. DL18 Equity OV

Standard Option Valuation				Page 1/2
MSFT	US	MICROSOFT CORP	Currency: USD	
Price of MSFT US Equity95			Hit 1 GO for save/send screen Hit 2 GO for notes Hit 3 GO for dividends Hit MENU for exotic option types Hit PAGE for scenario graph	
Strike:	95	100.000% (USD)	Rate:	4.742% Semiannual
Exercise Type:	European			
Put or Call:	Call			
Time to Expiration:	90	05:09	Model Type:	1 Default
Trade:	9/11/99	15:52		
Expiration:	12/10/99	21:02		
Settle Date:	9/11/99			
Exercise Delay:	0			
Option Valuation and Risk Parameters			Dividends	
Value	Percent	Time Value:	7.26294	Dividend Yield 0.00%
Price:	7.262938	Theta:	0.04277	Ex-Date Amount
Volatility:	35.821%	Premium:	7.64520	No dividends proj.
Delta:	0.56123	Parity:	0.00000	
Gamma:	0.02330	Gearing:	13.08011	
Vega:	0.18619	Rho:	0.11433	
Copyright 1999 BLOOMBERG L.P. Frankfurt 89-920410 Hong Kong 2-977-6000 London 171-330-7500 New York 212-318-2000 Princeton 609-279-3000 Singapore 226-3000 Sydney 2-9777-8686 Tokyo 3-3201-8900 Sao Paulo 11-3048-4500 1741-53-0 11-Sep-99 15:52:51				

Bloomberg

Figure 2.6 Bloomberg option valuation screen, call. Source: Bloomberg L.P.

premium you had to pay for the option. To adjust for the original cost of the option, sometimes one plots a diagram such as that shown in Figure 2.11. In this **profit diagram** for a call option I have subtracted from the payoff the premium originally paid for the call option. This figure is helpful because it shows how far into the money the asset must be at expiry before the option becomes profitable. The asset value marked S^* is the point

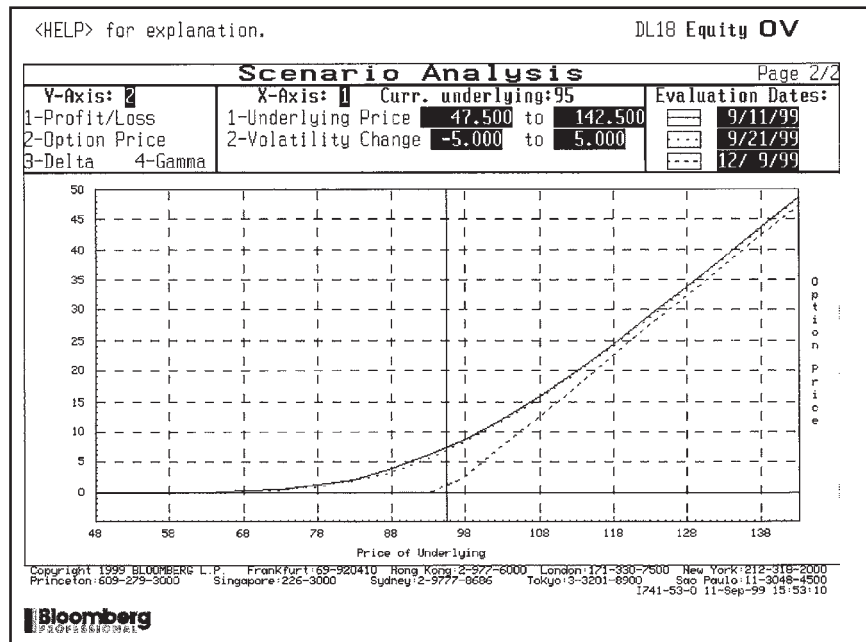


Figure 2.7 Bloomberg scenario analysis, call. Source: Bloomberg L.P.

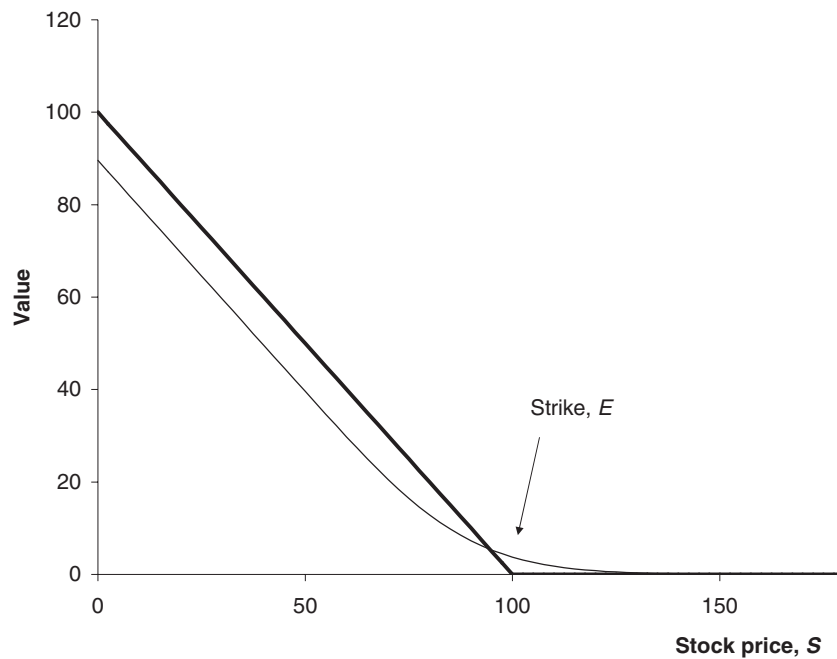


Figure 2.8 Payoff diagram for a put option.

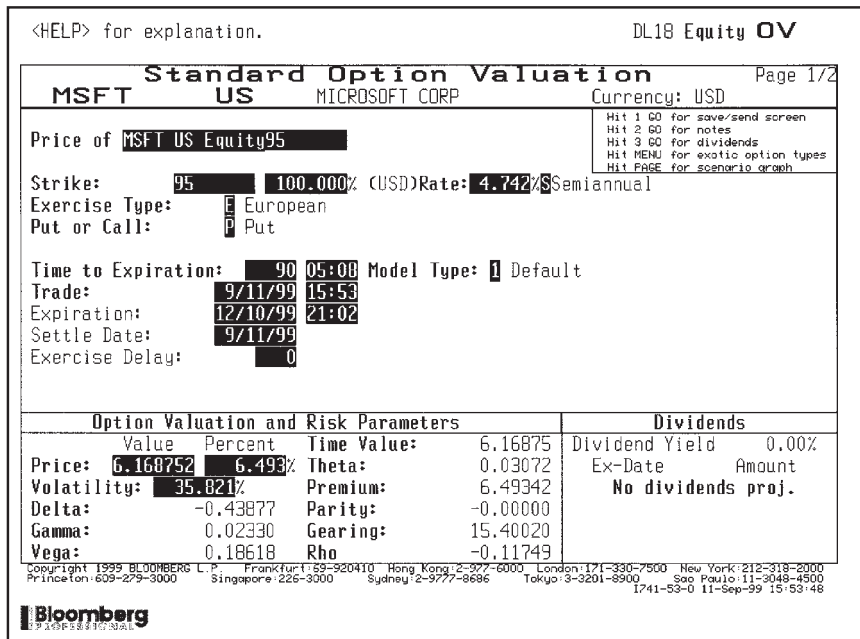


Figure 2.9 Bloomberg option valuation screen, put. Source: Bloomberg L.P.

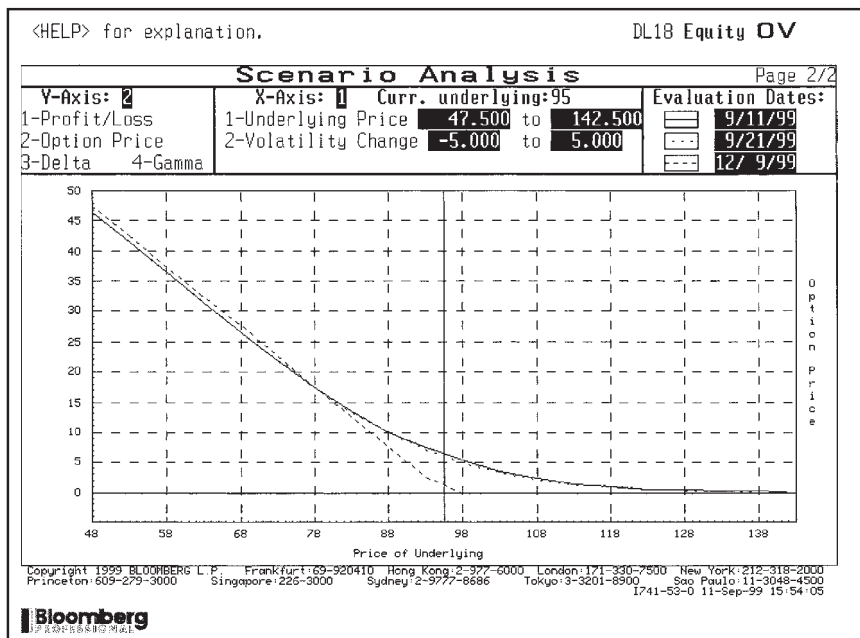


Figure 2.10 Bloomberg scenario analysis, put. Source: Bloomberg L.P.

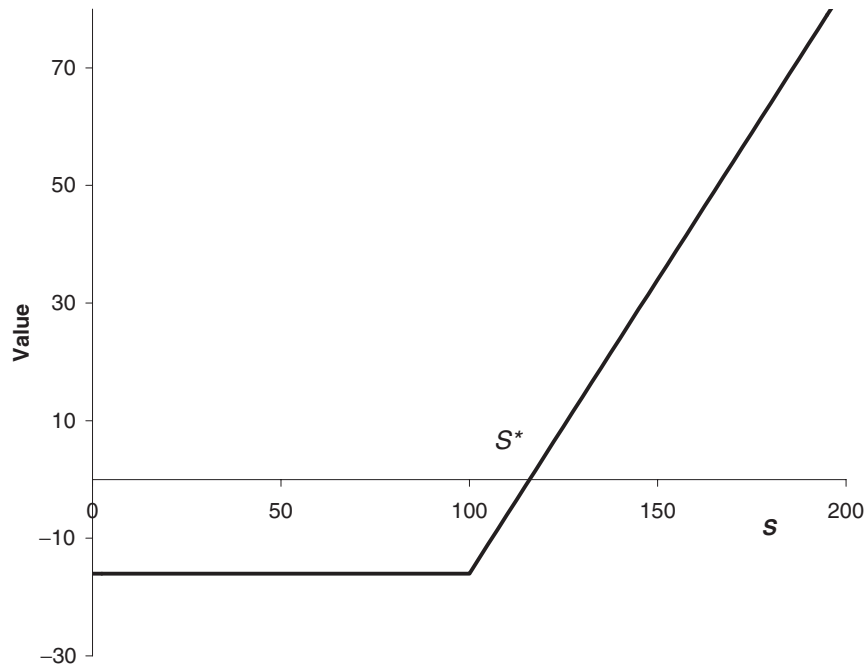


Figure 2.11 Profit diagram for a call option.

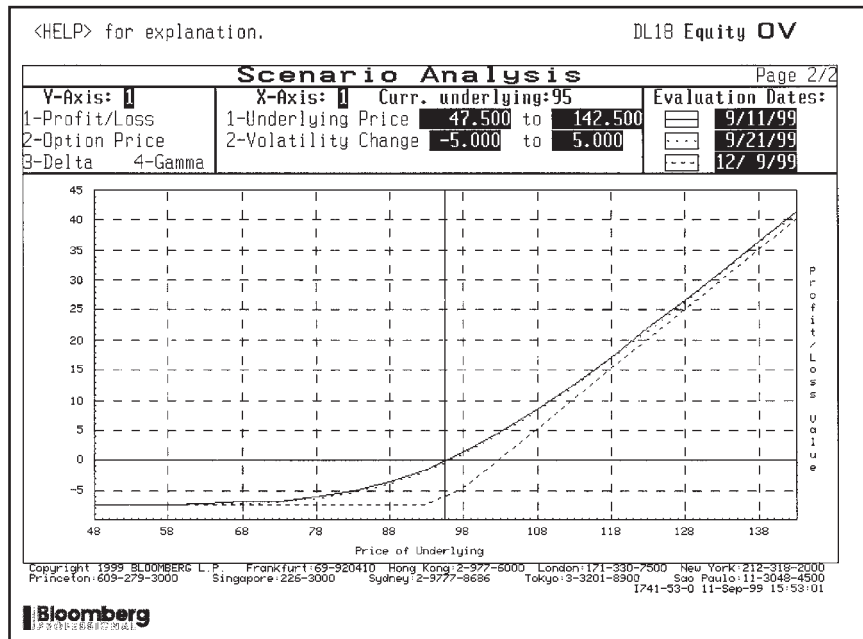


Figure 2.12 Profit diagram for a call. Source: Bloomberg L.P.

which divides profit from loss; if the asset at expiry is above this value then the contract has made a profit, if below the contract has made a loss.

As it stands, this profit diagram takes no account of the time value of money. The premium is paid up front but the payoff, if any, is only received at expiry. To be consistent one should either discount the payoff by multiplying by $e^{-r(T-t)}$ to value everything at the present, or multiply the premium by $e^{r(T-t)}$ to value all cashflows at expiry.

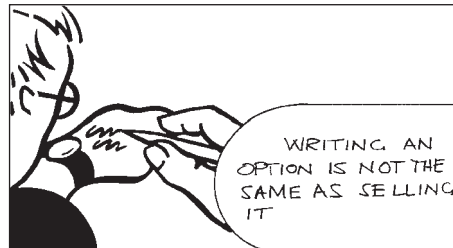
Figure 2.12 shows Bloomberg's call option profit diagram. Note that the profit today is zero; if we buy the option and immediately sell it we make neither a profit nor a loss (this is subject to issues of transaction costs).

2.5 WRITING OPTIONS

I have talked above about the rights of the purchaser of the option. But for every option that is sold, someone somewhere must be liable if the option is exercised. If I hold a call option entitling me to buy a stock some time in the future, who do I buy this stock from? Ultimately, the stock must be delivered by the person who **wrote** the option. The **writer** of an option is the person who promises to deliver the underlying asset, if the option is a call, or buy it, if the option is a put. The writer is the person who receives the premium.

In practice, most simple option contracts are handled through an exchange so that the purchaser of an option does not know who the writer is. The holder of the option can even sell the option on to someone else via the exchange to close his position. However, regardless of who holds the option, or who has handled it, the writer is the person who has the obligation to deliver or buy the underlying.

The asymmetry between owning and writing options is now clear. The purchaser of the option hands over a premium in return for special rights, and an uncertain outcome. The writer receives a guaranteed payment up front, but then has obligations in the future.



2.6 MARGIN

Writing options is very risky. The downside of buying an option is just the initial premium, the upside may be unlimited. The upside of writing an option is limited, but the downside could be huge. For this reason, to cover the risk of default in the event of an unfavorable outcome, the **clearing houses** that register and settle options insist on the deposit of a margin by the writers of options. Clearing houses act as counterparty to each transaction. Margin was described in Chapter 1.



2.7 MARKET CONVENTIONS

Most of the simpler options contracts are bought and sold through exchanges. These exchanges make it simpler and more efficient to match buyers with sellers. Part of

this simplification involves the conventions about such features of the contracts as the available strikes and expiries. For example, simple calls and puts come in **series**. This refers to the strike and expiry dates. Typically a stock has three choices of expiries trading at any time. Having standardized contracts traded through an exchange promotes liquidity of the instruments.

Some options are an agreement between two parties, often brought together by an intermediary. These agreements can be very flexible and the contract details do not need to satisfy any conventions. Such contracts are known as **over the counter** or **OTC** contracts. I give an example at the end of this chapter.

2.8 THE VALUE OF THE OPTION BEFORE EXPIRY

We have seen how much calls and puts are worth at expiry, and drawn these values in payoff diagrams. The question that we can ask, and the question that is central to this book, is ‘How much is the contract worth *now*, before expiry?’ How much would you pay for a contract, a piece of paper, giving you rights in the future? You may have no idea what the stock price will do between now and expiry in six months, say, but clearly the contract has value. At the very least you know that there is no downside to owning the option, the contract gives you specific rights but no *obligations*. Two things are clear about the contract value before expiry: the value will depend on how high the asset price is today and how long there is before expiry.

The higher the underlying asset today, the higher we might expect the asset to be at expiry of the option and therefore the more valuable we might expect a call option to be. On the other hand a put option might be cheaper by the same reasoning.

The dependence on time to expiry is more subtle. The longer the time to expiry, the more time there is for the asset to rise or fall. Is that good or bad if we own a call option? Furthermore, the longer we have to wait until we get any payoff, the less valuable will that payoff be simply because of the time value of money.

I will ask you to suspend disbelief for the moment (it won’t be the first time in the book) and trust me that we will be finding a ‘fair value’ for these options contracts. The aspect of finding the ‘fair value’ that I want to focus on now is the dependence on the asset price and time. I am going to use V to mean the value of the option, and it will be a function of the value of the underlying asset S at time t . Thus we can write $V(S, t)$ for the value of the contract.

We know the value of the contract *at expiry*. If I use T to denote the expiry date then at $t = T$ the function V is known, it is just the payoff function. For example if we have a call option then

$$V(S, T) = \max(S - E, 0).$$

This is the function of S that I plotted in the earlier payoff diagrams. Now I can tell you what the fine lines are in Figures 2.5 and 2.8, they are the values of the contracts $V(S, t)$ *at some time before expiry*, plotted against S . I have not specified how long before expiry, since the plot is for explanatory purposes only.

Time Out...

Functions of two variables

The option value is a function of two variables, asset price S and time t . If it helps, think of V as being the height of a mountain with the two variables being distances in the northerly and westerly directions. Later we're going to be looking at the slope of this mountain in each of the two directions... these will be sensitivities of the option price to changes in the asset and in time. These slopes or gradients are what you experience in your car when you see a sign such as '1-in-10 gradient.' That is precisely the same as a slope of 0.1.

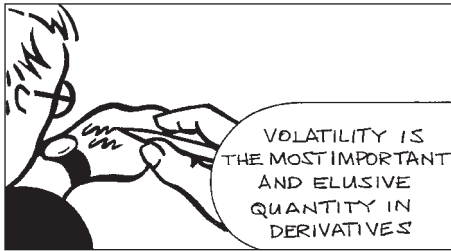


2.9 **FACTORS AFFECTING DERIVATIVE PRICES**

The two most important factors affecting the prices of options are the value of the underlying asset S and the time to expiry t . These quantities are **variables** meaning that they inevitably change during the life of the contract; if the underlying did not change then the pricing would be trivial. This contrasts with the **parameters** that affect the price of options.

Examples of parameters are the interest rate and strike price. The interest rate will have an effect on the option value via the time value of money since the payoff is received in the future. The interest rate also plays another role which we will see later. Clearly the strike price is important, the higher the strike in a call, the lower the value of the call.

If we have an equity option then its value will depend on any dividends that are paid on the asset during the option's life. If we have an FX option then its value will depend on the interest rate received by the foreign currency.



There is one important parameter that I have not mentioned, and which has a major impact on the option value. That parameter is the **volatility**. Volatility is a measure of the amount of fluctuation in the asset price, a measure of the randomness. Figure 2.13 shows two simulated asset price paths, the more jagged of the two has the higher volatility. The technical definition of volatility is the 'annualized standard deviation of the asset returns.' I will show how to

measure this parameter in Chapter 4.

Volatility is a particularly interesting parameter because it is so hard to estimate. And having estimated it, one finds that it never stays constant and is unpredictable.

The distinction between parameters and variables is very important. I shall be deriving equations for the value of options, partial differential equations. These equations will involve differentiation with respect to the variables, but the parameters, as their name suggests, remain as parameters in the equations.



Volatility and timescales

Time Out...

Volatility

Remember our first coin-tossing experiment back in Chapter 1? Try this again, but instead of multiplying by a factor of 1.01 or 0.99, use factors of 1.02 and 0.98. Now plot the time series. This is an example of a more volatile path. If you're feeling strong, try the following experiment.

Play around with different scale factors, 1.01 and 0.99, 1.02 and 0.98, 1.05 and 0.95, keeping them symmetric about 1 to start with. Now try different 'timescales,' i.e. toss the coin only once every two units of time, then once every four units. Would you call a path with large but infrequent moves as volatile as one with smaller but more frequent moves?

2.10 SPECULATION AND GEARING

If you buy a far out-of-the-money option it may not cost very much, especially if there is not very long until expiry. If the option expires worthless, then you also haven't lost very

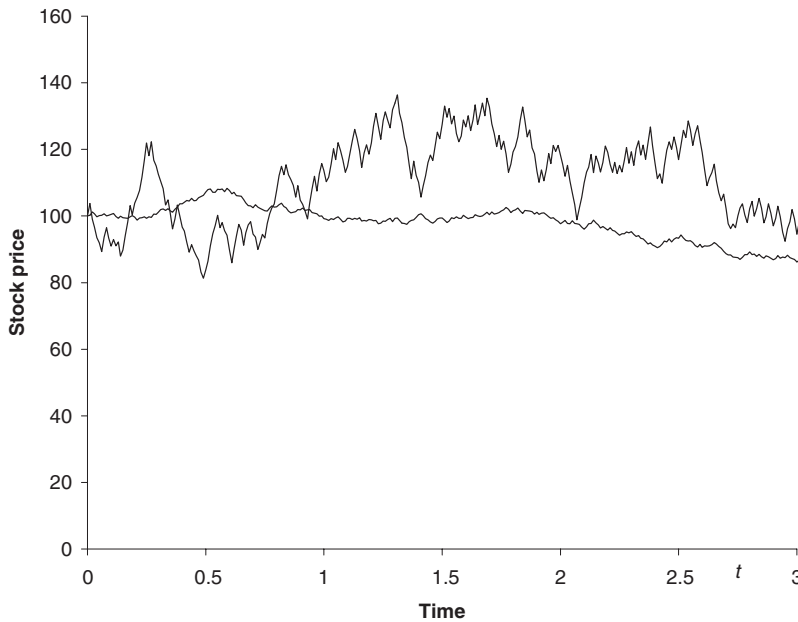


Figure 2.13 Two (simulated) asset price paths, one is much more volatile than the other.

much. However, if there is a dramatic move in the underlying, so that the option expires in the money, you may make a large profit relative to the amount of the investment. Let me give an example.

Example Today's date is 14th April and the price of Wilmott Inc. stock is \$666. The cost of a 680 call option with expiry 22nd August is \$39. I expect the stock to rise significantly between now and August, how can I profit if I am right?

Buy the stock Suppose I buy the stock for \$666. And suppose that by the middle of August the stock has risen to \$730. I will have made a profit of \$64 per stock. More importantly my investment will have risen by

$$\frac{730 - 666}{666} \times 100 = 9.6\%.$$

Buy the call If I buy the call option for \$39, then at expiry I can exercise the call, paying \$680 to receive something worth \$730. I have paid \$39 and I get back \$50. This is a profit of \$11 per option, but in percentage terms I have made

$$\frac{\text{value of asset at expiry} - \text{strike} - \text{cost of call}}{\text{cost of call}} \times 100 = \frac{730 - 680 - 39}{39} \times 100 = 28\%.$$

This is an example of **gearing** or **leverage**. The out-of-the-money option has a high gearing, a possible high payoff for a small investment. The downside of this leverage is that the call option is more likely than not to expire completely worthless and you will lose all of your investment. If Wilmott Inc. remains at \$666 then the stock investment has the same value but the call option experiences a 100% loss.

Highly leveraged contracts are very risky for the writer of the option. The buyer is only risking a small amount; although he is very likely to lose, his downside is limited to his initial premium. But the writer is risking a large loss in order to make a probable small profit. The writer is likely to think twice about such a deal unless he can offset his risk by buying other contracts. This offsetting of risk by buying other related contracts is called **hedging**.

Gearing explains one of the reasons for buying options. If you have a strong view about the direction of the market then you can exploit derivatives to make a better return, if you are right, than buying or selling the underlying.

2.11 EARLY EXERCISE

The simple options described above are examples of **European options** because exercise is only permitted *at expiry*. Some contracts allow the holder to exercise *at any time* before expiry, and these are called **American options**. American options give the holder more rights than their European equivalent and can therefore be more valuable, and they can never be less valuable. The main point of interest with American-style contracts is deciding *when* to exercise. In Chapter 3 I will discuss American options, and show how to determine when it is *optimal* to exercise, so as to give the contract the highest value.

Note that the terms 'European' and 'American' do not in any way refer to the continents on which the contracts are traded.

Finally, there are **Bermudan options**. These allow exercise on specified dates, or in specified periods. In a sense they are half way between European and American since exercise is allowed on some days and not on others.



2.12 PUT-CALL PARITY

Imagine that you buy one European call option with a strike of E and an expiry of T and that you write a European put option with the same strike and expiry. Today's date is t . The payoff you receive at T for the call will look like the line in the first plot of Figure 2.14. The payoff for the put is the line in the second plot in the figure. Note that the sign of the payoff is negative,

you *wrote* the option and are liable for the payoff. The payoff for the portfolio of the two options is the sum of the individual payoffs, shown in the third plot. The payoff for this portfolio of options is

$$\max(S(T) - E, 0) - \max(E - S(T), 0) = S(T) - E,$$

where $S(T)$ is the value of the underlying asset at time T .

The right-hand side of this expression consists of two parts, the asset and a fixed sum E . Is there another way to get exactly this payoff? If I buy the asset today it will cost me $S(t)$ and be worth $S(T)$ at expiry. I don't know what the value $S(T)$ will be but I do know how to guarantee to get that amount, and that is to buy the asset.

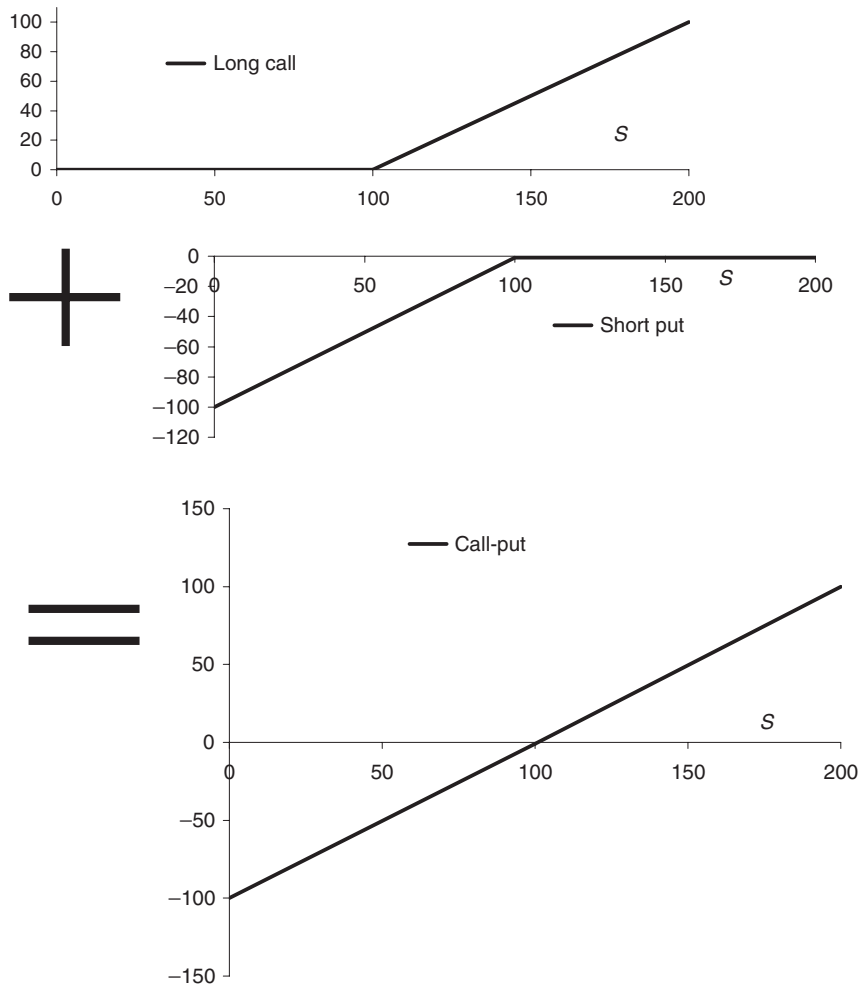


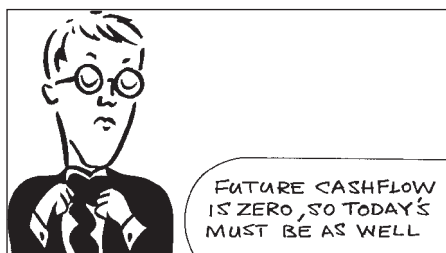
Figure 2.14 Schematic diagram showing put-call parity.

What about the E term? To lock in a payment of E at time T involves a cash flow of $Ee^{-r(T-t)}$ at time t . The conclusion is that the portfolio of a long call and a short put gives me exactly the same payoff as a long asset, short cash position. The equality of these cashflows is independent of the future behavior of the stock and is model independent:

$$C - P = S - Ee^{-r(T-t)},$$

where C and P are today's values of the call and the put respectively. This relationship holds at any time up to expiry and is known as **put-call parity**. If this relationship did not hold then there would be riskless arbitrage opportunities.

In Table 2.1 I show the cashflows in the perfectly hedged portfolio. In this table I have set up the cashflows to have a guaranteed value of zero at expiry.

**Table 2.1** Cashflows in a hedged portfolio of options and asset.

Holding	Worth today (t)	Worth at expiry (T)
Call	C	$\max(S(T) - E, 0)$
– Put	$-P$	$-\max(E - S(T), 0)$
– Stock	$-S(t)$	$-S(T)$
Cash	$Ee^{-r(T-t)}$	E
Total	$C - P - S(t) + Ee^{-r(T-t)}$	0



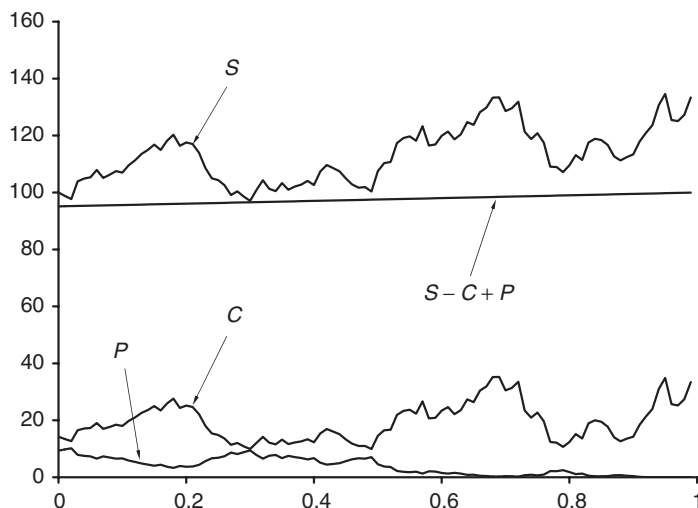
See how this might work in practice on the CD

Time Out...

A simulation of put-call parity

Below are four plots, all with time along the horizontal axis. The first is of some asset price. The second is the value of a call option on that asset. You don't need to know details of the contract, such as strike and expiry. Nor do you need to know how I calculated the value.

The third plot is of a put option (same strike and expiry as the call, whatever they were). The fourth plot is stock value minus call value plus put value. Observe how it grows exponentially, just like cash in the bank. This is a graphical illustration of put-call parity.



2.13 BINARYS OR DIGITALS

The original and still most common contracts are the vanilla calls and puts. Increasingly important are the **binary** or **digital options**. These contracts have a payoff at expiry that is discontinuous in the underlying asset price. An example of the payoff diagram for one of these options, a **binary call**, is shown in Figure 2.15. This contract pays \$1 at expiry, time T , if the asset price is then greater than the exercise price E . Again, and as with the rest of the figures in this chapter, the bold line is the payoff and the fine line is the contract value some time before expiry.

Why would you invest in a binary call? If you think that the asset price will rise by expiry, to finish above the strike price then you might choose to buy either a vanilla call or a binary call. The vanilla call has the best upside potential, growing linearly with S beyond the strike. The binary call, however, can never pay off more than the \$1. If you expect the underlying to rise dramatically then it may be best to buy the vanilla call. If you believe that the asset rise will be less dramatic then buy the binary call. The gearing of the vanilla call is greater than that for a binary call if the move in the underlying is large.

Figure 2.16 shows the payoff diagram for a **binary put**, the holder of which receives \$1 if the asset is *below* E at expiry. The binary put would be bought by someone expecting a modest fall in the asset price.

There is a particularly simple binary put-call parity relationship. What do you get at expiry if you hold both a binary call and a binary put with the same strikes and expiries? The answer is that you will always get \$1 regardless of the level of the underlying at expiry. Thus

$$\text{Binary call} + \text{Binary put} = e^{-r(T-t)}.$$

What would the table of cashflows look like for the perfectly hedged digital portfolio?

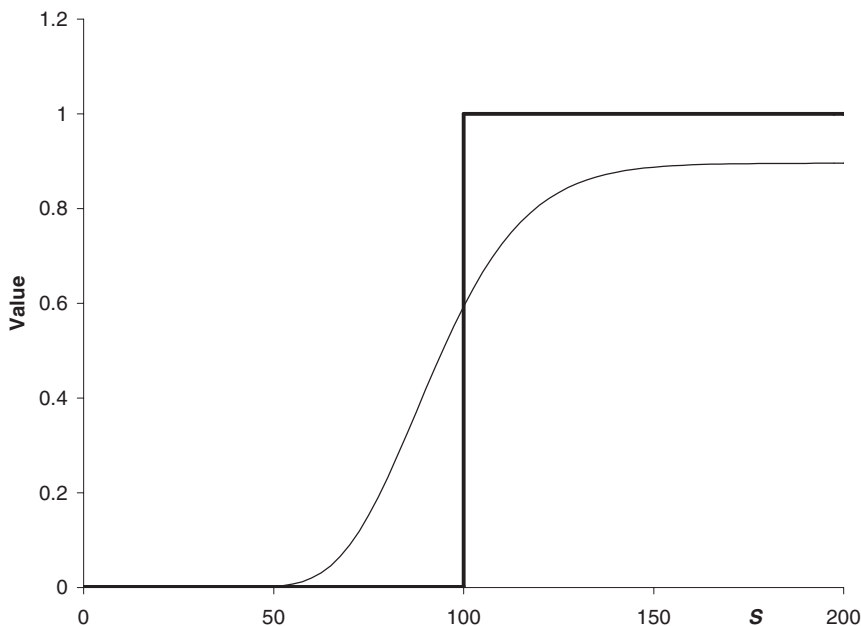


Figure 2.15 Payoff diagram for a binary call option.

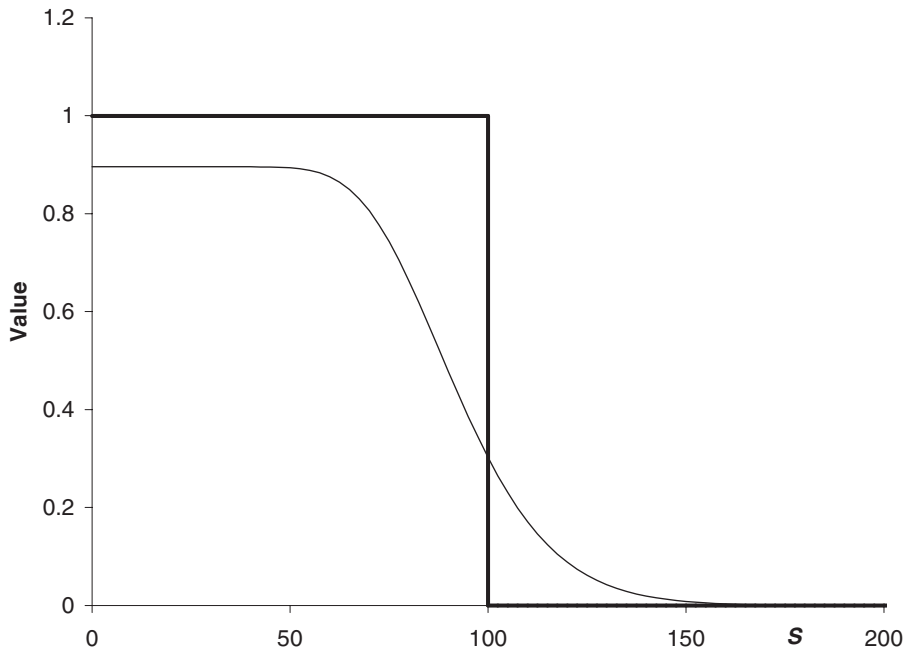


Figure 2.16 Payoff diagram for a binary put option.

2.14 BULL AND BEAR SPREADS

A payoff that is similar to a binary option can be made up with vanilla calls. This is our first example of a **portfolio of options** or an **option strategy**.

Suppose I buy one call option with a strike of 100 and write another with a strike of 120 and with the same expiration as the first then my resulting portfolio has a payoff that is shown in Figure 2.17. This payoff is zero below 100, 20 above 120 and linear in between. The payoff is continuous, unlike the binary call, but has a payoff that is superficially similar. This strategy is called a **bull spread** (or a **call spread**) because it benefits from a bull, i.e. rising, market.

The payoff for a general bull spread, made up of calls with strikes E_1 and E_2 , is given by

$$\frac{1}{E_2 - E_1} (\max(S - E_1, 0) - \max(S - E_2, 0)),$$

where $E_2 > E_1$. Here I have bought/sold $(E_2 - E_1)^{-1}$ of each of the options so that the maximum payoff is scaled to 1.

If I write a put option with strike 100 and buy a put with strike 120 I get the payoff shown in Figure 2.18. This is called a **bear spread** (or a **put spread**), benefiting from a bear, i.e. falling, market. Again, it is very similar to a binary put except that the payoff is continuous.

Because of put-call parity it is possible to build up these payoffs using other contracts. A strategy involving options of the same type (i.e. calls or puts) is called a **spread**.

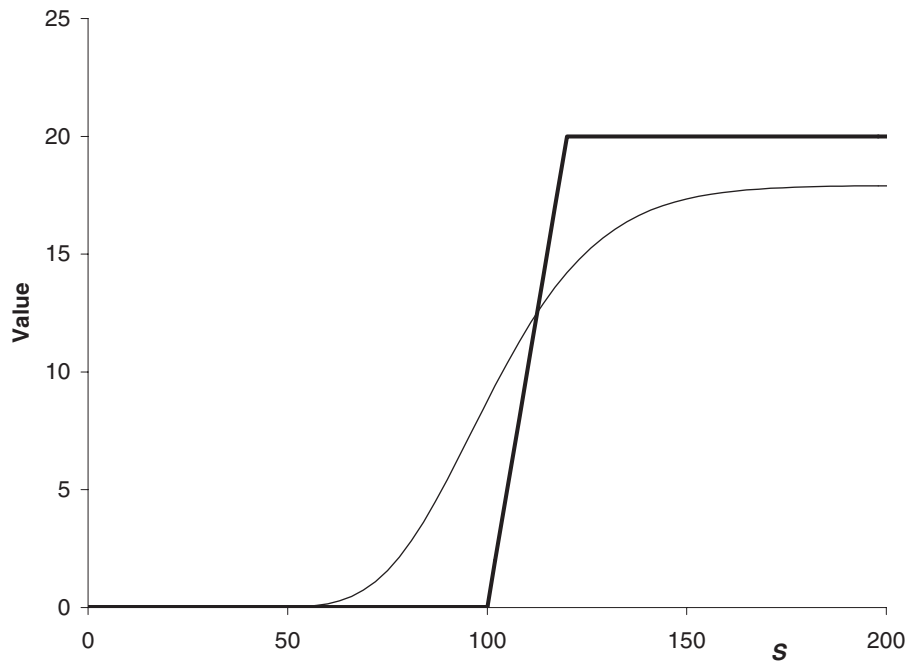


Figure 2.17 Payoff diagram for a bull spread.

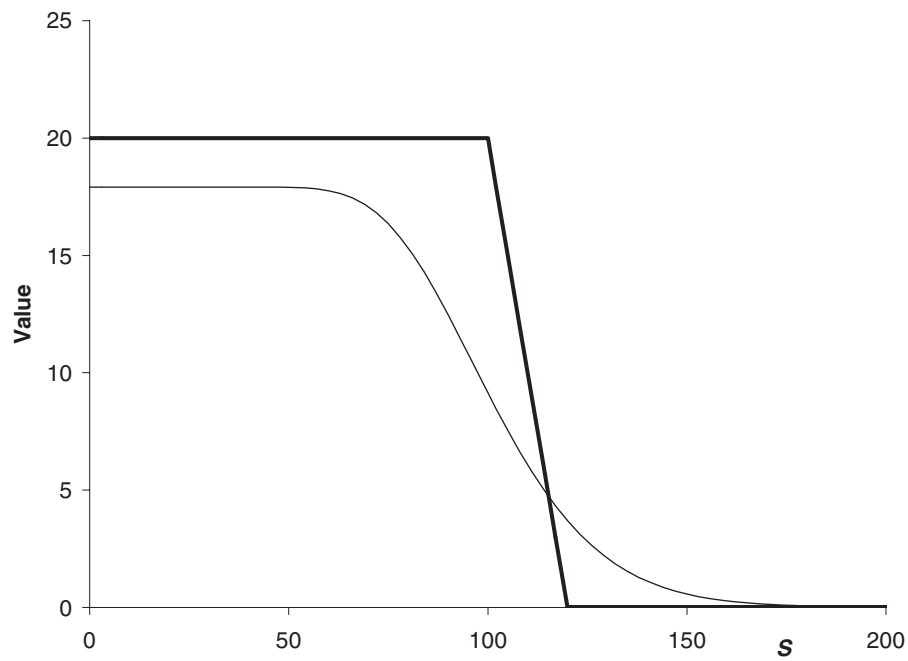


Figure 2.18 Payoff diagram for a bear spread.

2.15 STRADDLES AND STRANGLES

If you have a precise view on the behavior of the underlying asset you may want to be precise in your choice of option; simple calls, puts, and binaries may be too crude.

The **straddle** consists of a call and a put with the same strike. The payoff diagram is shown in Figure 2.19. Such a position is usually bought at the money by someone who expects the underlying to either rise or fall, but not to remain at the same level. For example, just before an anticipated major news item stocks often show a 'calm before the storm.' On the announcement the stock suddenly moves either up or down depending on whether or not the news was favorable to the company. They may also be bought by technical traders who see the stock at a key support or resistance level and expect the stock to either break through dramatically or bounce back.

The straddle would be sold by someone with the opposite view, someone who expects the underlying price to remain stable.

Figure 2.20 shows the Bloomberg screen for setting up a straddle. Figure 2.21 shows the profit and loss for this position at various times before expiry. The profit/loss is the option value less the upfront premium.

The **strangle** is similar to the straddle except that the strikes of the put and the call are different. The contract can be either an **out-of-the-money strangle** or an **in-the-money strangle**. The payoff for an out-of-the money strangle is shown in Figure 2.22. The motivation behind the purchase of this position is similar to that for the purchase of a straddle. The difference is that the buyer expects an even larger move in the underlying one way or the other. The contract is usually bought when the asset is around the middle of the two strikes and is cheaper than a straddle. This cheapness means that the gearing

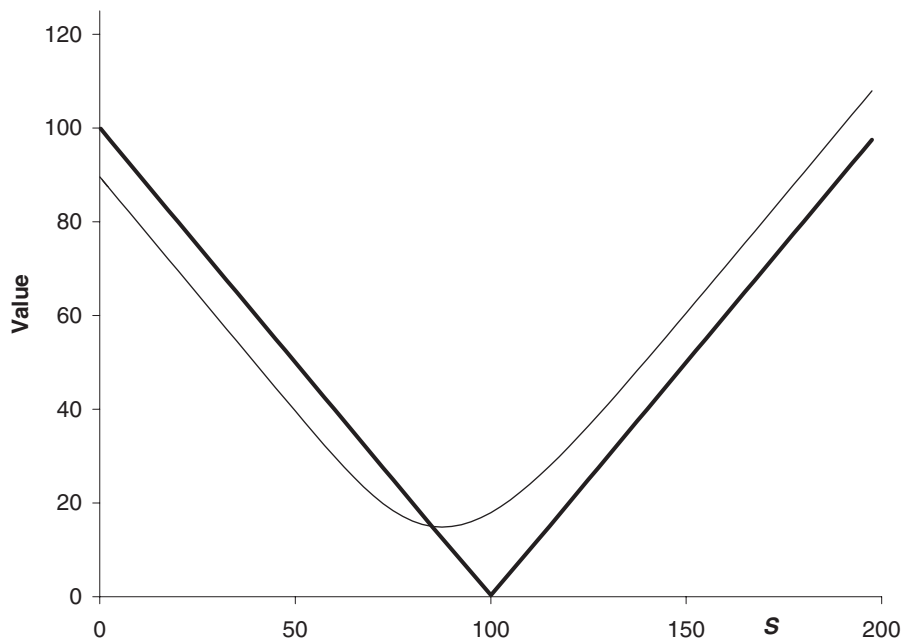


Figure 2.19 Payoff diagram for a straddle.

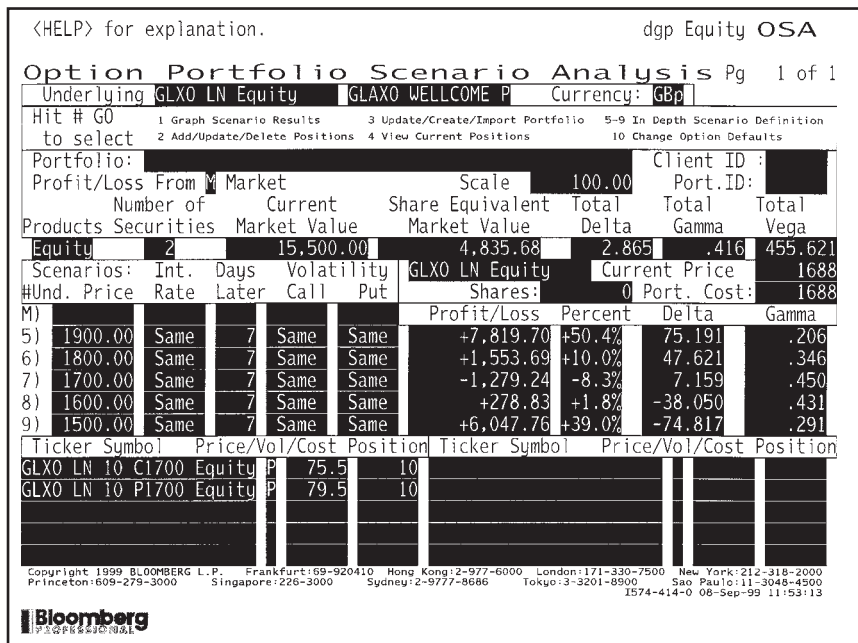


Figure 2.20 A portfolio of two options making up a straddle. Source: Bloomberg L.P.

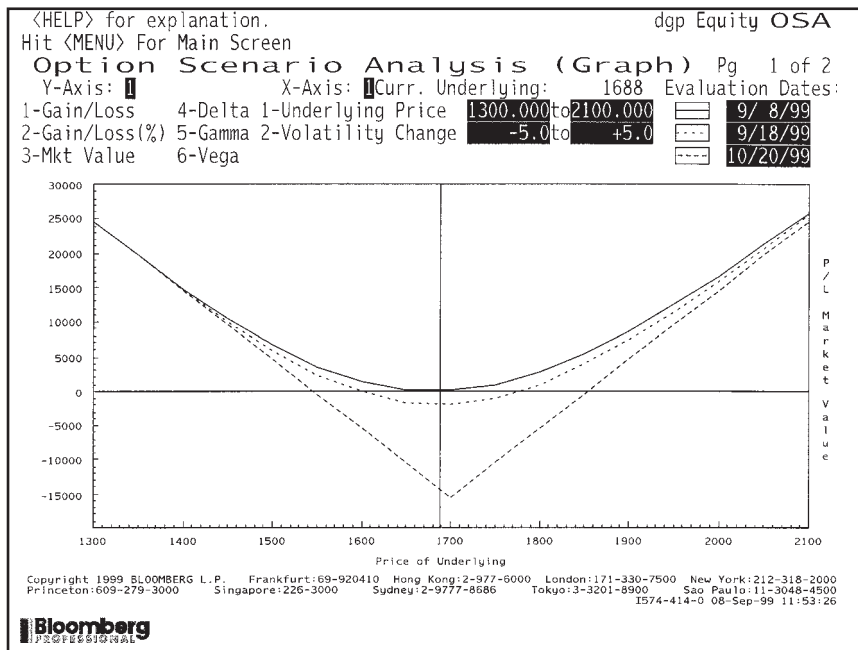


Figure 2.21 Profit/loss for the straddle at several times before expiry. Source: Bloomberg L.P.

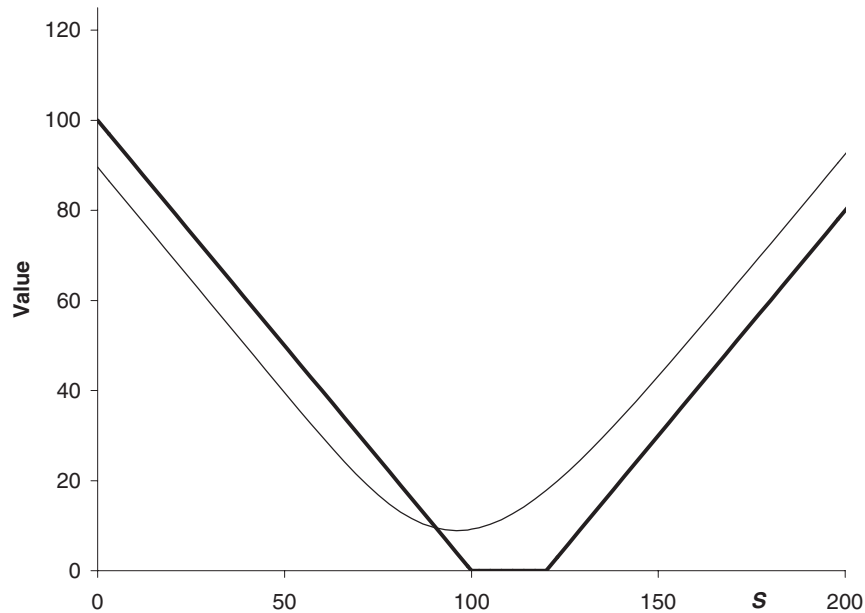


Figure 2.22 Payoff diagram for a strangle.

for the out-of-the-money strangle is higher than that for the straddle. The downside is that there is a much greater range over which the strangle has no payoff at expiry, for the straddle there is only the one point at which there is no payoff.

There is another reason for a straddle or strangle trade that does not involve a view on the direction of the underlying. These contracts are bought or sold by those with a view on the direction of volatility, they are one of the simplest **volatility trades**. Because of the relationship between the price of an option and the volatility of the asset one can speculate on the direction of volatility. Do you expect the volatility to rise? If so, how can you benefit from this? Until we know more about this relationship, we cannot go into this in more detail.

Straddles and strangles are rarely held until expiry.

A strategy involving options of different types (i.e. both calls and puts) is called a **combination**.

2.16 RISK REVERSAL

The **risk reversal** is a combination of a long call, with a strike above the current spot, and a short put, with a strike below the current spot. Both have the same expiry. The payoff is shown in Figure 2.23.

The risk reversal is a very special contract, popular with practitioners. Its value is usually quite small between the strikes and related to the market's expectations of the behavior of volatility.

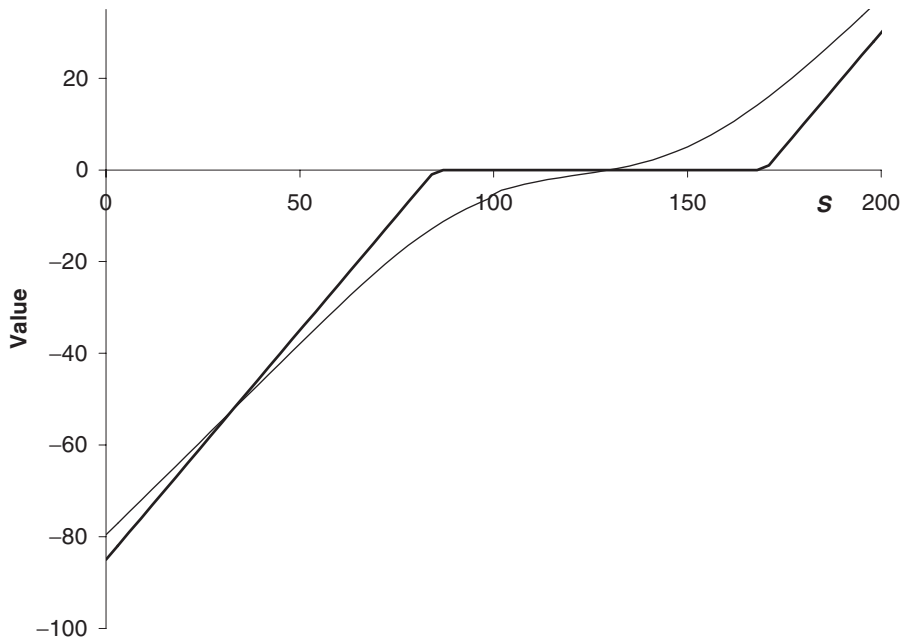


Figure 2.23 Payoff diagram for a risk reversal.

2.17 BUTTERFLIES AND CONDORS

A more complicated strategy involving the purchase and sale of options with *three* different strikes is a **butterfly spread**. Buying a call with a strike of 90, writing two calls struck at 100 and buying a 110 call gives the payoff in Figure 2.24. This is the kind of position you might enter into if you believe that the asset is not going anywhere, either up or down. Because it has no large upside potential (in this case the maximum payoff is 10) the position will be relatively cheap. With options, cheap is good.

The **condor** is like a butterfly except that four strikes, and four call options, are used. The payoff is shown in Figure 2.25.

2.18 CALENDAR SPREADS

All of the strategies I have described above have involved buying or writing calls and puts with different strikes *but all with the same expiration*. A strategy involving options with different expiry dates is called a **calendar spread**. You may enter into such a position if you have a precise view on the timing of a market move as well as the direction of the move. As always the motive behind such a strategy is to reduce the payoff at asset values and times which you believe are irrelevant, while increasing the payoff where you think it will matter. Any reduction in payoff will reduce the overall value of the option position.

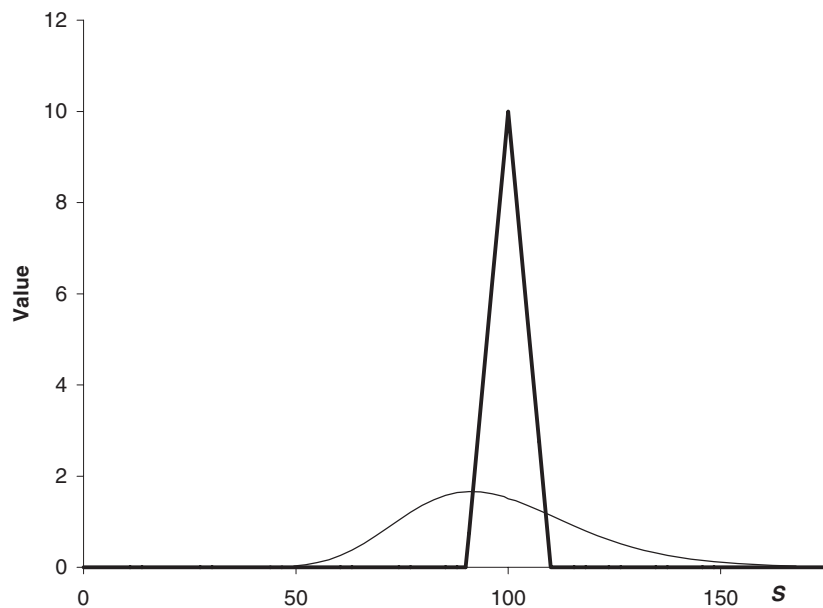


Figure 2.24 Payoff diagram for a butterfly spread.

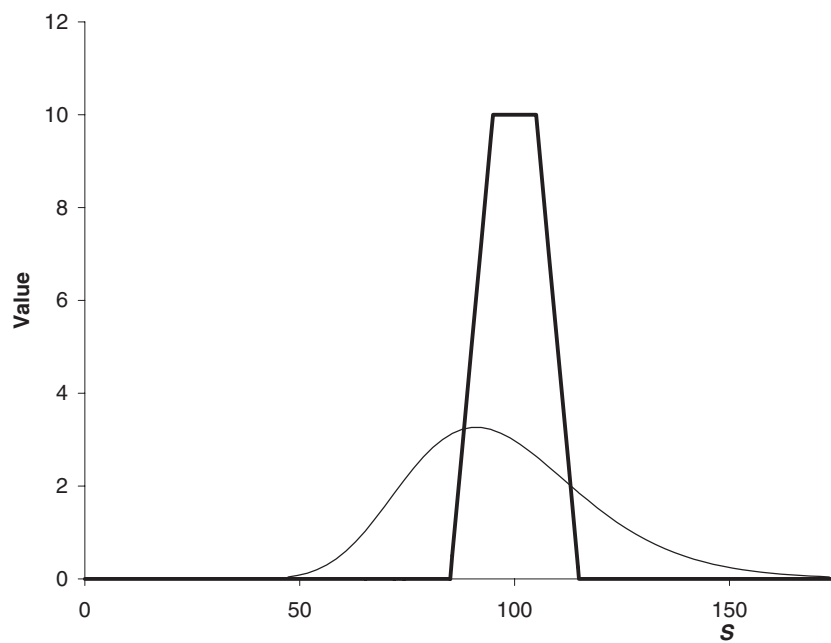


Figure 2.25 Payoff diagram for a condor.

2.19 LEAPS AND FLEX

LEAPS or **long-term equity anticipation securities** are longer-dated exchange-traded calls and puts. They began trading on the CBOE in the late 1980s. They are standardized so that they expire in January each year and are available with expiries up to three years. They come with three strikes, corresponding to at the money and 20% in and out of the money with respect to the underlying asset price when issued.

Figure 2.26 shows LEAPS quoted in *The Wall Street Journal Europe*.

In 1993 the CBOE created **FLEX** or **FLexible EXchange-traded options** on several indices. These allow a degree of customization, in the expiry date (up to five years), the strike price and the exercise style.

2.20 WARRANTS

A contract that is very similar to an option is a **warrant**. Warrants are call options issued by a company on its own equity. The main differences between traded options and warrants are the timescales involved, warrants usually have a longer lifespan, and on exercise the company issues new stock to the warrant holder. On exercise, the holder of a *traded* option receives stock that has already been issued. Exercise is usually allowed any time before expiry, but after an initial waiting period.

The typical lifespan of a warrant is five or more years. Occasionally **perpetual warrants** are issued, these have no maturity.

2.21 CONVERTIBLE BONDS

Convertible bonds or **CBs** have features of both bonds and warrants. They pay a stream of coupons with a final repayment of principal at maturity, but they can be converted into the underlying stock before expiry. On conversion rights to future coupons are lost. If the

LEAPS-LONG TERM									
DJ INDUS AVG - CB									
Dec 01	104	p	42	7 $\frac{3}{4}$	+	3 $\frac{1}{4}$		475	
Dec 01	140	p	44	22 $\frac{1}{4}$	+	2 $\frac{1}{2}$		106	
S & P 100 INDEX - CB									
Dec 01	140	c	5	38	+	6		105	
S & P 500 INDEX - CB									
Dec 00	70	p	60	7 $\frac{1}{16}$	+	1 $\frac{1}{8}$		8993	
Dec 00	90	p	5	13 $\frac{1}{16}$	+	1 $\frac{1}{2}$		15789	
Dec 00	100	p	4	11 $\frac{5}{16}$	+	1 $\frac{1}{2}$		18242	
Dec 00	110	p	85	23 $\frac{1}{4}$	+	1 $\frac{1}{4}$		15068	
Dec 00	112 $\frac{1}{2}$	p	1	3	+	3 $\frac{1}{8}$		7496	
Dec 00	115	p	22	3 $\frac{3}{8}$	+	5 $\frac{1}{8}$		19282	
Dec 00	117 $\frac{1}{2}$	p	10	7 $\frac{1}{8}$	+	1 $\frac{3}{8}$		839	
Dec 00	120	p	52	4 $\frac{3}{8}$	+	5 $\frac{1}{8}$		17787	
Dec 00	125	p	6	5 $\frac{1}{8}$	+	5 $\frac{1}{8}$		5809	
Dec 00	130	p	18	6 $\frac{1}{4}$	+	7 $\frac{1}{8}$		7322	
Dec 00	140	p	90	9	+	1 $\frac{1}{4}$		10159	
Dec 00	145	p	2	9 $\frac{1}{2}$	+	3 $\frac{1}{4}$		935	
Call Volume			39	Open Int			6,469,087		
Put Volume			20	Open Int			4,671,720		

Figure 2.26 *The Wall Street Journal Europe* of 5th January 2000, LEAPS. Reproduced by permission of Dow Jones & Company, Inc.

stock price is low then there is little incentive to convert to the stock, the coupon stream is more valuable. In this case the CB behaves like a bond. If the stock price is high then conversion is likely and the CB responds to the movement in the asset. Because the CB can be converted into the asset, its value has to be at least the value of the asset. This makes CBs similar to American options; early exercise and conversion are mathematically the same.

2.22 OVER THE COUNTER OPTIONS

Not all options are traded on an exchange. Some, known as over the counter or OTC options, are sold privately from one counterparty to another. In Figure 2.27 is the term

*Preliminary and Indicative
For Discussion Purposes Only*

Over-the-counter Option linked to the S&P500 Index

Option Type	European put option, with contingent premium feature
Option Seller	XXXX
Option Buyer	[dealing name to be advised]
Notional Amount	USD 20MM
Trade Date	[]
Expiration Date	[]
Underlying Index	S&P500
Settlement	Cash settlement
Cash Settlement Date	5 business days after the Expiration Date
Cash Settlement Amount	Calculated as per the following formula: $\# \text{Contracts} * \max[0, S\&P_{\text{strike}} - S\&P_{\text{final}}]$ where $\# \text{Contracts} = \text{Notional Amount} / S\&P_{\text{initial}}$ This is the same as a conventional put option: S&Pstrike will be equal to 95% of the closing price on the Trade Date S&Pfinal will be the level of the Underlying Index at the valuation time on the Expiration Date S&Pinitial is the level of the Underlying Index at the time of execution
Initial Premium Amount	[2%] of Notional Amount
Initial Premium Payment Date	5 business days after Trade Date
Additional Premium Amounts	[1.43%] of Notional Amount per Trigger Level
Additional Premium Payment Dates	The Additional Premium Amounts shall be due only if the Underlying Index at any time from and including the Trade Date and to and including the Expiration Date is equal to or greater than any of the Trigger Levels.
Trigger Levels	103%, 106% and 109% of S&P500initial
Documentation	ISDA
Governing law	New York

This indicative term sheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 2.27 Term sheet for an OTC ‘put.’

sheet for an OTC put option, having some special features. A **term sheet** specifies the precise details of an OTC contract. In this OTC put the holder gets a put option on S&P500, but more cheaply than a vanilla put option. This contract is cheap because part of the premium does not have to be paid until and unless the underlying index trades above a specified level. Each time that a new level is reached an extra payment is triggered. This feature means that the contract is not vanilla, and makes the pricing more complicated. We will be discussing special features like the ones in this contract in later chapters. Quantities in square brackets will be set at the time that the deal is struck.

2.23 SUMMARY

We now know the basics of options and markets, and a few of the simplest trading strategies. We know some of the jargon and the reasons why people might want to buy an option. We've also seen another example of no arbitrage in put-call parity. This is just the beginning. We don't know how much these instruments are worth, how they are affected by the price of the underlying, how much risk is involved in the buying or writing of options. And we have only seen the very simplest of contracts; there are many, many more complex products to examine. All of these issues are going to be addressed in later chapters.

FURTHER READING

- McMillan (1996) and Options Institute (1995) describe many option strategies used in practice.
- Most exchanges have websites. The London International Financial Futures Exchange website contains information about the money markets, bonds, equities, indices and commodities. See www.liffe.com. For information about options and derivatives generally, see www.cboe.com, the Chicago Board Options Exchange website. The American Stock Exchange is on www.amex.com and the New York Stock Exchange on www.nyse.com.
- Derivatives have often had bad press (and there's probably more to come). See Miller (1997) for a discussion of the pros and cons of derivatives.
- The best books on options are Hull (2005) and Cox & Rubinstein (1985), modesty forbids me mentioning others. (Oh, all right then, *PWOQF2*.)

EXERCISES

1. Find the value of the following portfolios of options at expiry, as a function of the share price:
 - (a) Long one share, long one put with exercise price E ,
 - (b) Long one call and one put, both with exercise price E ,
 - (c) Long one call, exercise price E_1 , short one call, exercise price E_2 , where $E_1 < E_2$,

- (d) Long one call, exercise price E_1 , long one put, exercise price E_2 . There are three cases to consider,
 - (e) Long two calls, one with exercise price E_1 and one with exercise price E_2 , short two calls, both with exercise price E , where $E_1 < E < E_2$.
2. What is the difference between a payoff diagram and a profit diagram? Illustrate with a portfolio of short one share, long two calls with exercise price E .
 3. A share currently trades at \$60. A European call with exercise price \$58 and expiry in three months trades at \$3. The three month default-free discount rate is 5%. A put is offered on the market, with exercise price \$58 and expiry in three months, for \$1.50. Do any arbitrage opportunities now exist? If there is a possible arbitrage, then construct a portfolio that will take advantage of it. (This is an application of put-call parity.)
 4. A three-month, 80 strike, European call option is worth \$11.91. The 90 call is \$4.52 and the 100 call is \$1.03. How much is the butterfly spread?
 5. Using the notation $V(E)$ to mean the value of a European call option with strike E , what can you say about $\frac{\partial V}{\partial E}$ and $\frac{\partial^2 V}{\partial E^2}$ for options having the same expiration?
- Hint: Consider call and butterfly spreads and the absence of arbitrage.